

Table 3: Performance of a Mask R-CNN R50 network in mAP@0.5-0.95 scores trained on DocLayNet with different class label sets. The reduced label sets were obtained by either down-mapping or dropping labels.

Class-count	11	6	5	4
Caption	68	Text	Text	Text
Footnote	71	Text	Text	Text
Formula	60	Text	Text	Text
List-item	81	Text	82	Text
Page-footer	62	62	-	-
Page-header	72	68	-	-
Picture	72	72	72	72
Section-header	68	67	69	68
Table	82	83	82	82
Text	85	84	84	84
Title	77	Sec.-h.	Sec.-h.	Sec.-h.
Overall	72	73	78	77

Learning Curve

One of the fundamental questions related to any dataset is if it is “large enough”. To answer this question for DocLayNet, we performed a data ablation study in which we evaluated a Mask R-CNN model trained on increasing fractions of the DocLayNet dataset. As can be seen in Figure 5, the mAP score rises sharply in the beginning and eventually levels out. To estimate the error-bar on the metrics, we ran the training five times on the entire data-set. This resulted in a 1% error-bar, depicted by the shaded area in Figure 5. In the inset of Figure 5, we show the exact same data-points, but with a logarithmic scale on the x-axis. As is expected, the mAP score increases linearly as a function of the data-size in the inset. The curve ultimately flattens out between the 80% and 100% mark, with the 80% mark falling within the error-bars of the 100% mark. This provides a good indication that the model would not improve significantly by yet increasing the data size. Rather, it would probably benefit more from improved data consistency (as discussed in Section 3), data augmentation methods [23], or the addition of more document categories and styles.

Impact of Class Labels

The choice and number of labels can have a significant effect on the overall model performance. Since PubLayNet, DocBank and DocLayNet all have different label sets, it is of particular interest to understand and quantify this influence of the label set on the model performance. We investigate this by either down-mapping labels into more common ones (e.g. *Caption*→*Text*) or excluding them from the annotations entirely. Furthermore, it must be stressed that all mappings and exclusions were performed on the data before model training. In Table 3, we present the mAP scores for a Mask R-CNN R50 network on different label sets. Where a label is down-mapped, we show its corresponding label, otherwise it was excluded. We present three different label sets, with 6, 5 and 4 different labels respectively. The set of 5 labels contains the same labels as PubLayNet. However, due to the different definition of

Table 4: Performance of a Mask R-CNN R50 network with document-wise and page-wise split for different label sets. Naive page-wise split will result in ~10% point improvement.

Class-count Split	11		5	
	Doc	Page	Doc	Page
Caption	68	83		
Footnote	71	84		
Formula	60	66		
List-item	81	88	82	88
Page-footer	62	89		
Page-header	72	90		
Picture	72	82	72	82
Section-header	68	83	69	83
Table	82	89	82	90
Text	85	91	84	90
Title	77	81		
All	72	84	78	87

lists in PubLayNet (grouped list-items) versus DocLayNet (separate list-items), the label set of size 4 is the closest to PubLayNet, in the assumption that the *List* is down-mapped to *Text* in PubLayNet. The results in Table 3 show that the prediction accuracy on the remaining class labels does not change significantly when other classes are merged into them. The overall macro-average improves by around 5%, in particular when *Page-footer* and *Page-header* are excluded.

Impact of Document Split in Train and Test Set

Many documents in DocLayNet have a unique styling. In order to avoid overfitting on a particular style, we have split the train-, test- and validation-sets of DocLayNet on document boundaries, i.e. every document contributes pages to only one set. To the best of our knowledge, this was not considered in PubLayNet or DocBank. To quantify how this affects model performance, we trained and evaluated a Mask R-CNN R50 model on a modified dataset version. Here, the train-, test- and validation-sets were obtained by a randomised draw over the individual pages. As can be seen in Table 4, the difference in model performance is surprisingly large: page-wise splitting gains ~10% in mAP over the document-wise splitting. Thus, random page-wise splitting of DocLayNet can easily lead to accidental overestimation of model performance and should be avoided.

Dataset Comparison

Throughout this paper, we claim that DocLayNet’s wider variety of document layouts leads to more robust layout detection models. In Table 5, we provide evidence for that. We trained models on each of the available datasets (PubLayNet, DocBank and DocLayNet) and evaluated them on the test sets of the other datasets. Due to the different label sets and annotation styles, a direct comparison is not possible. Hence, we focussed on the common labels among the datasets. Between PubLayNet and DocLayNet, these are *Picture*,